

Pre-Algebra Practice Test IV-Version Two

Instructions

- i) As always, write a title, name, date, and course.
- ii) Number each problem, circle the number, and show all work.
- iii) Draw a rectangle around your answers.
- iv) Rewrite the equation whenever you perform an operation on both sides to preserve the result of the previous step.

① Graph the following ordered pairs. Label your graph properly.

A (75, -25)

B (100, 0)

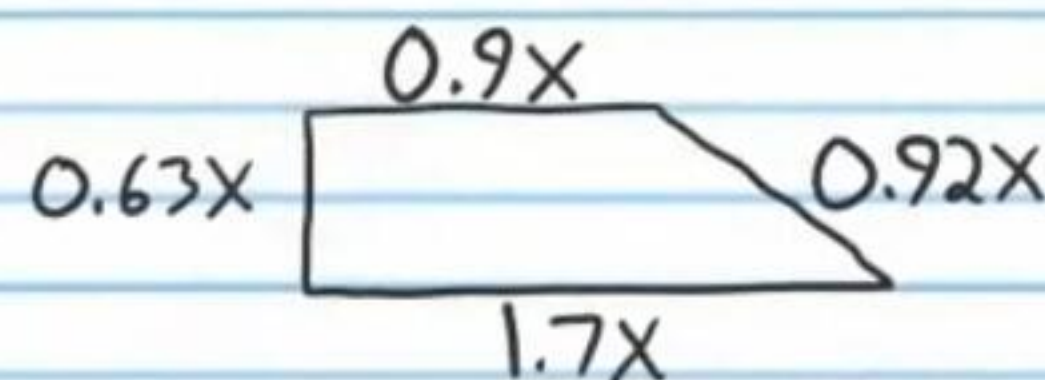
C (0, -75)

D (-50, -50)

E (0, 0)

② 203% of what number is 194.88?

③ Find the perimeter of the trapezoid below.



④ You loan a friend \$ 265 and charge him 1% simple interest paid daily. Over time the debt grows to \$280.90. How much time passed?

⑤ The sum of three consecutive even integers is 318. Identify these integers.

⑥ 55 is what percent of 75? Round to the nearest tenth of a percent.

Solve. Round to the nearest thousandth.

⑦ $3.11x + 2 = 5.6 - 8x$

⑧ Over time, a star will lose mass and eventually die. If a star's mass decreases by 2%, its new mass is 1.5 nonillion kilograms (a very large number). What was its mass before the decrease? Round to the nearest thousandth.

⑨ Find the average of the following numbers: $2y, y, 3y, 3y$.

⑩ Find the algebraic solution and the graphical solution of the inequality below.

$$-2(x-4) - 7 < 1 + x$$

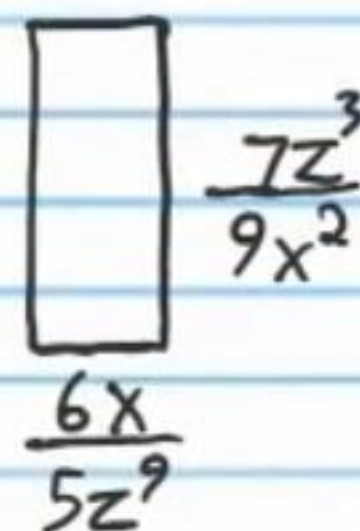
⑪ Find the volume of a cone with base radius $2x^2$ and height $6x^3$. Leave π as a factor in your answer.

⑫ Solve.

$$-7.6y - 4y + 2y = 12 - 0.6y$$

⑬ Find the area of a trapezoid with bases $7z$ and $3z$, and height $5z^3$.

⑭ Find the area of the rectangle below.



Solve.

⑮ $2(-2.1x + 8) - (5x - 2) = -0.2x$

⑯ Evaluate your answer for problem #14 if $x = -2$ and $z = 1$.

⑰ Use the simple interest formula to find the amount of money in a savings account if $P = \$450$, $r = 0.09$, and $t = 10$.

⑱ If the area of a rectangle is 64 ft^2 and the width is 11 ft , find the length.

⑲ If 18 is increased to 36.7, what is the percent increase? Round to the nearest tenth of a percent.

Find the algebraic solutions and the graphical solutions of the following inequalities.

⑳ $-6x + 2 \geq -9x$

㉑ $-5y + 9y - 4 < y + 2$

Graph the following equations. Show all work and identify the points you use to graph the equations. Label your graphs properly.

$$(22) \quad \frac{x}{5} - 9 = y$$

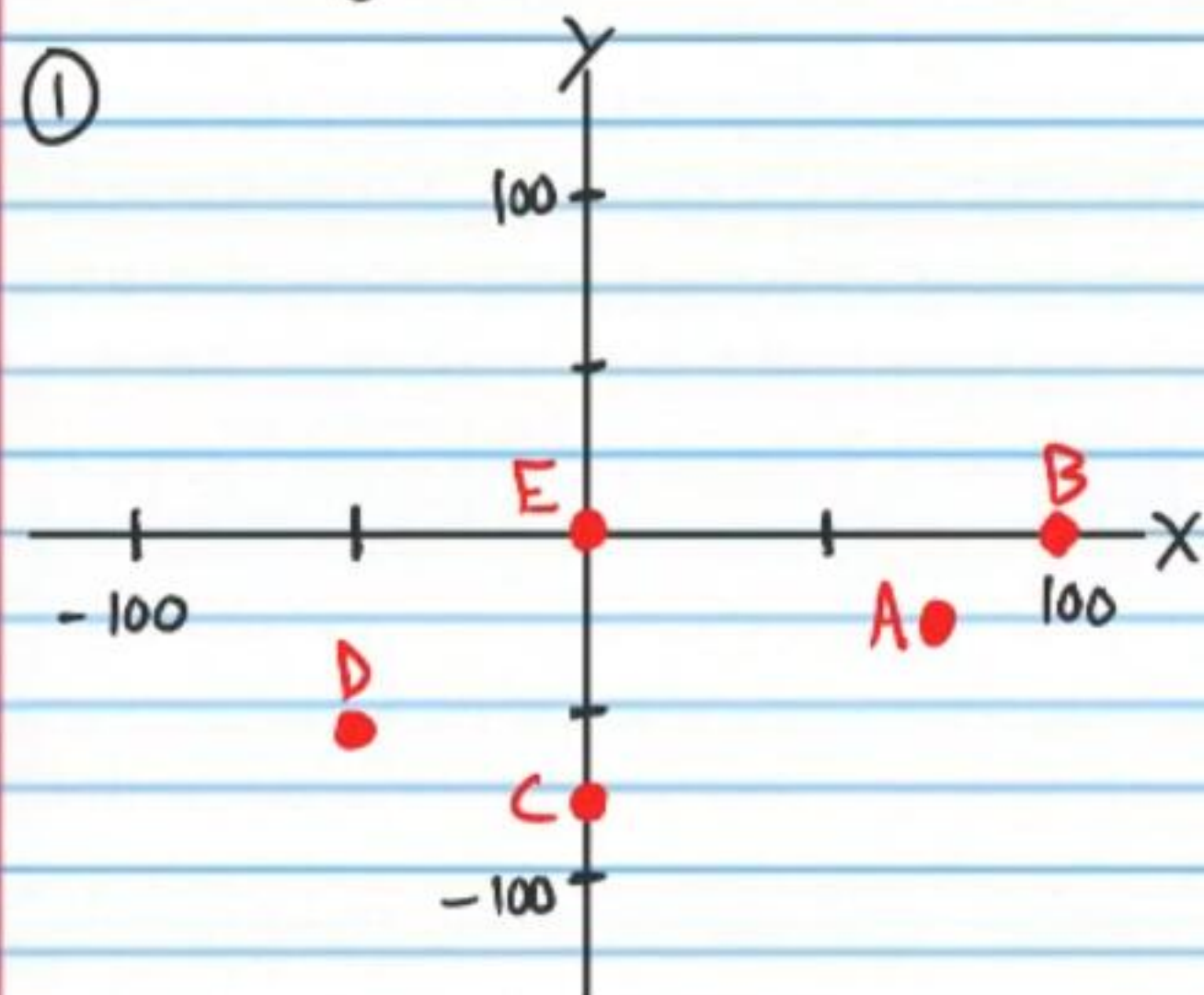
$$(23) \quad x + y = -4$$

Graph the following inequalities. Show all work and identify the points you use to graph the inequalities. Label your graphs properly.

$$(24) \quad x < 3y + 50$$

$$(25) \quad 5x + 6y \geq 2$$

Pre-Algebra Practice Test IV - Version Two Answers



② 96

③ $4.15x$

④ 6 days

⑤ 104, 106, 108

⑥ 73.3%

⑦ $x \approx 0.324$

⑧ 1.531 nonillion kg

⑨ $9\frac{1}{4}$

⑩ $x > 0$

⑪ $V = 8\pi x^7$

⑫ $y = -\frac{4}{3}$

⑬ $A = 25z^4$

⑭ $A = \frac{14}{15xz^6}$

⑮ $x = 2$

⑯ $A = -\frac{7}{15}$

⑰ \$855

⑱ $L = 5\frac{9}{11} \text{ ft}$

⑲ 103.9%

⑳ $x \geq -\frac{2}{3}$



㉑ $y < 2$

